ANALYTICS PORTFOLIO · INDEPENDENT PROJECT · OCTOBER 2026

Late Orders, Lost Stars

Payments and delivery analytics on 99,092 real marketplace orders: what drives bad reviews, how slow payments make orders late, and how to prove a fix with an experiment.

+51.8 pp
chance of a 1–2 star review when an order arrives late, after controls

+29%
late-delivery risk when payment approval takes over 24 hours

3,567
orders per arm for the designed A/B test, validated by 2,000 simulations

THE QUESTION

Which payment and fulfilment problems cost a marketplace its customers' trust, and how would we prove that a fix works?

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Data & Business Analytics

BUILT WITH

SQL · SQLite | Python · pandas | statsmodels | scikit-learn | YAML metric layer | Claude (AI-assisted)

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Data: Olist, Brazilian E-Commerce Public Dataset [1] · 99,092 orders, 3,029 sellers, Jan 2017 – Aug 2018. Canadian context: Payments Canada, Canadian Payment Methods and Trends Report 2025 [3]. Full references on page 10.

EXECUTIVE SUMMARY

Late delivery is the problem to fix, and payments are part of the cause

Across 99,092 orders and R$13.4M in GMV, 6.8% of deliveries arrived after the promised date and 14.6% of reviews were 1–2 stars. Five results shape the recommendations.

+51.8 pp

A late delivery is the strongest driver of bad reviews

1–2 star rate goes from 9.2% on time to 62.4% late, and 79.3% at 8+ days late. Controlled for state, category, value and month.

33.1h

Slow payment methods eat into the delivery window

Boleto takes 33.1 hours to approve vs 4.4 for cards. Orders approved after 24 hours are 29% more likely to arrive late.

34%

Installments carry the high-ticket orders

7+ installment orders are 16.2% of card orders but 34% of card GMV. They also explain 91.6% of payment reconciliation breaks.

29.4%

A small seller group drives a large share of the damage

172 at-risk sellers handle 17.4% of orders but 29.4% of late deliveries and 24.7% of 1–2 star reviews.

0.68

Late orders can be flagged at payment approval

Out-of-time AUC 0.68; the top 30% risk orders hold 52.4% of late deliveries, which makes a targeted experiment possible.

RECOMMENDATION

Start the delivery clock at payment approval, run a **proactive delay-alert A/B test** on high-risk orders (7,134 orders, about 16 weeks), and put the 172 at-risk sellers on a fulfilment plan.

REPORTED AS FOUND

Late delivery does **not** significantly change 180-day repurchase ($p = 0.233$). Repeat buying is rare here (3.2%), so reviews are the outcome to manage.

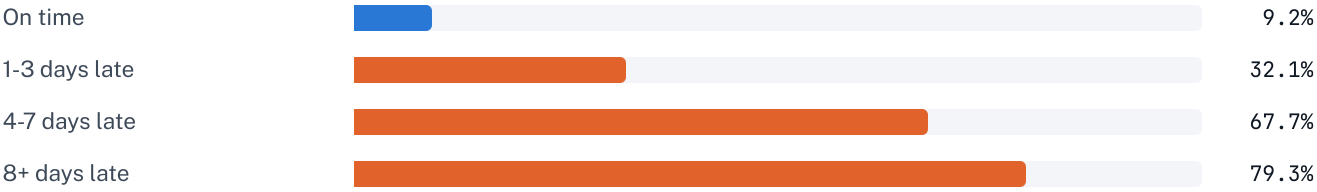
FINDING 1 · CUSTOMER EXPERIENCE

A late delivery is the strongest driver of bad reviews

On-time orders get a 1–2 star review 9.2% of the time. Late orders get one 62.4% of the time.

Share of reviews scored 1–2 stars, by days past the promised date

Delivered orders with a review, Jan 2017 – Aug 2018 · n = 96,203



DELAY	ORDERS	AVG STARS	1–2 STAR RATE
On time	89,672	4.29	9.2%
1-3 days late	1,869	3.29	32.1%
4-7 days late	1,802	2.10	67.7%
8+ days late	2,860	1.70	79.3%

+51.8 pp

effect of a late delivery on the chance of a 1–2 star review (95% CI 50.5–53), OLS (statsmodels [5]) with state, category, value and month fixed effects, robust SE, n = 95,560

4.29 → 2.27

average review score, on-time vs late orders

NULL RESULT

New customers whose first order was late came back within 180 days slightly less often (2.78% vs 3.2%), but the controlled difference of -0.35 pp is not significant (p = 0.233). The study does not claim an effect on repurchase.

So what: because the effect is large and rises with every day of delay, even a partial fix to lateness moves reviews. The next question is what makes orders late in the first place.

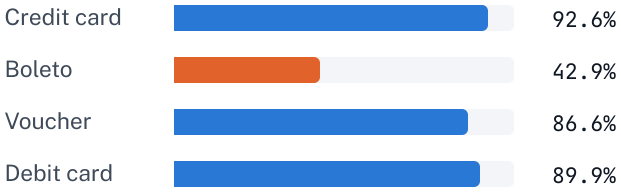
FINDING 2 · PAYMENTS

Slow payment approval makes orders late

> **The delivery promise is set at purchase, so a payment that takes a day or more to approve comes straight out of the shipping window.**

Approved within 24 hours

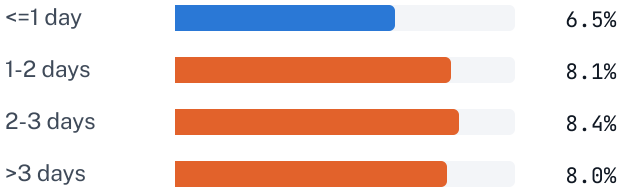
By primary payment method



Avg approval: boleto 33.1h · card 4.4h. Boleto is a bank slip paid in cash or online banking.

Late-delivery rate by approval time

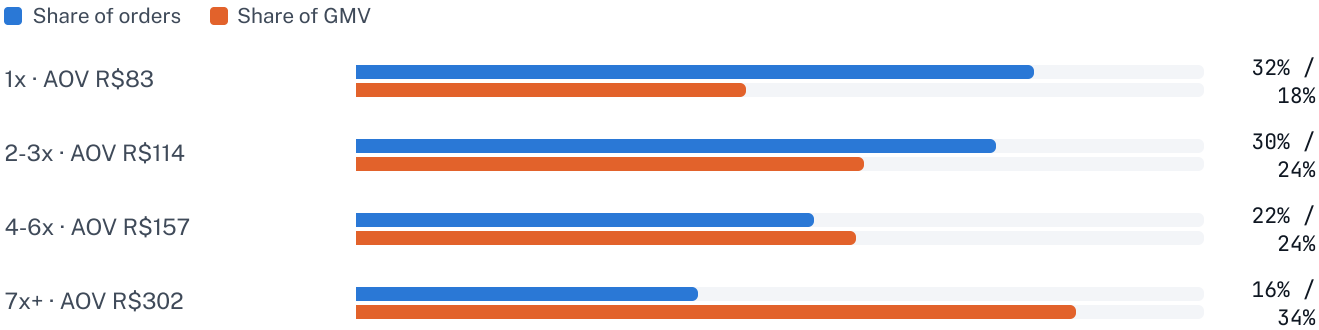
Hours from purchase to payment approval



Adjusted effect of approval > 24h: +1.86 pp (95% CI 1.42–2.3) on a 6.5% base.

Installment plans carry the high-ticket orders

Credit-card orders by number of installments



RECONCILIATION

Payment totals matched items plus freight on all but **249** of 98,665 orders (0.25%). **228** of those breaks (91.6%) were installment card orders where the buyer paid interest on top; boleto had 0.

FINDING 3 · SELLERS AND COMPLAINTS

A small group of sellers drives a large share of the damage

> **The top 10% of sellers hold 67.4% of GMV. Of 633 sellers with 30+ orders, 172 run late or poorly reviewed at 1.5× the platform rate or worse.**

17.4%
of orders come from at-risk sellers

29.4%
of late deliveries come from them

24.7%
of 1–2 star reviews come from them

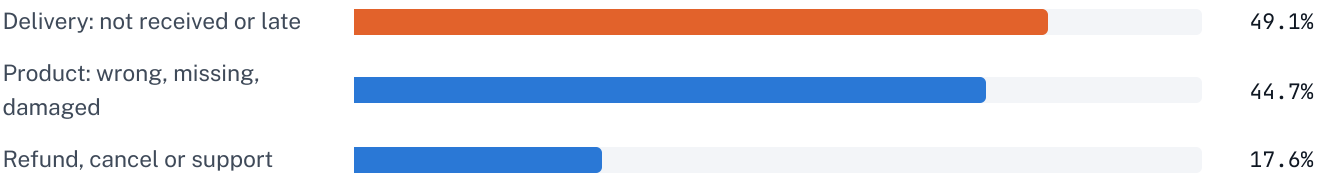
At-risk sellers vs other established sellers

Sellers with 30+ orders · rule: late rate or 1–2 star rate at 1.5× the platform average or worse

GROUP	SELLERS	MEDIAN ORDERS	LATE RATE	1–2 STAR RATE	AVG STARS	GMV
At-risk sellers	**172**	**62**	**11.8%**	**23.2%**	**3.76**	**R$2.3M**
Other sellers (30+ orders)	461	72	4.5%	11.9%	4.20	R$8.0M

What 1–2 star reviews complain about

10,737 comments in Portuguese · keyword tagging, shares corrected with held-out precision and recall



TAG QUALITY ON 60 HAND-LABELLED HELD-OUT REVIEWS	PRECISION	RECALL	RAW SHARE
Delivery issue	0.79	0.68	42.3%
Product issue	1.00	0.65	29.1%

PAYMENTS RISK SIGNAL

Among 1–2 star reviews on orders recorded as **delivered on time**, 26.5% still describe a delivery problem, often “marked delivered but never received”. On a payments platform these are the orders most likely to become item-not-received disputes and chargebacks.

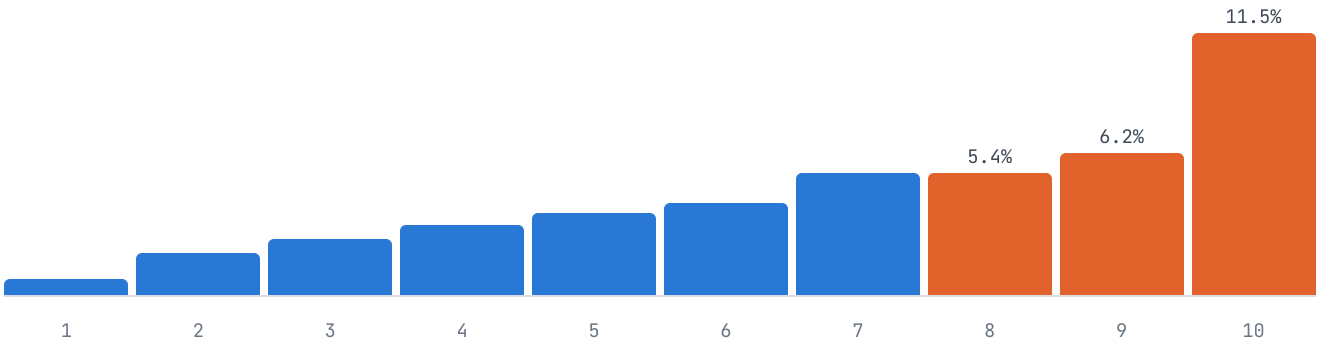
MODEL · OUT-OF-TIME VALIDATION

Late orders can be flagged at payment approval

> **Trained on orders to March 2018 and tested on April–August 2018, the model reaches AUC 0.68. The top decile is late 11.5% of the time, 2.6× the average.**

Actual late-delivery rate by predicted risk decile

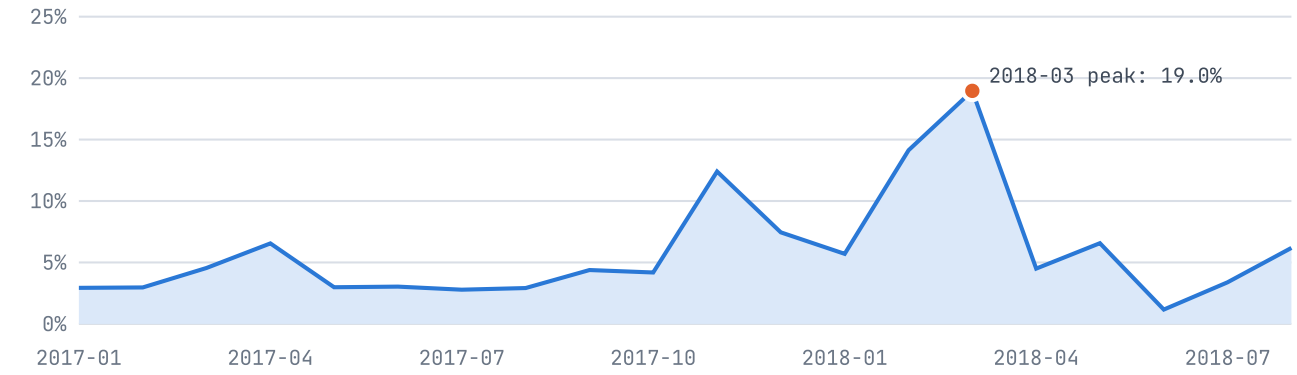
Test set: 32,150 delivered orders · decile 10 = highest predicted risk



Top 3 deciles hold 52.4% of late orders: the experiment's target group

Monthly late-delivery rate

Share of delivered orders that missed the promised date



ALGORITHM	Histogram gradient boosting (scikit-learn [6]), categorical features native
FEATURES	Known at payment approval only: customer and seller state, same-state flag, category, payment method, installments, approval lag, promised days, item value, freight, item and seller counts, month, weekday
WHY OUT-OF-TIME	A random split would leak seasonal patterns; testing on later months shows how the model would perform after launch

EXPERIMENT DESIGN

Proactive delay alerts for high-risk orders

> **Setting a realistic ETA and alerting customers before a slip lowers 1-2 star reviews among orders the risk model flags as likely late.**

ELIGIBILITY	orders scored in the top 3 risk deciles at payment approval (these held 52.4% of late orders out-of-time)
RANDOMIZATION	customer (hash of customer_unique_id) so repeat buyers stay in one arm
TREATMENT	Realistic ETA at approval, an alert before a predicted slip, and a seller escalation; control keeps the current experience
PRIMARY METRIC	1–2 star review rate, from the metric registry · baseline 11.1% from 9,589 historical orders
EFFECT TO DETECT	−2 pp (18% relative) · α 0.05, power 0.8, two-sided
SAMPLE AND TIME	**3,567 orders per arm**, 7,134 total · about 16 weeks at 2018 volume
GUARDRAILS	cancel_rate · avg delivery days · support contacts per order · alert cost per order · sample ratio mismatch (chi-square p < 0.001 = stop)
DECISION RULE	ship if low_review_rate falls at p < 0.05 and no guardrail worsens by more than 2%
RISKS	review lag: reviews arrive after delivery, so read results 30 days after enrollment closes · interference: sellers serve both arms; escalations may also help control orders (bias toward null) · novelty: run whole weeks and check the effect by week

Sample size by detectable effect

Two-sided, α 0.05, power 0.80

DROP	REL.	PER ARM	WEEKS
-1.0 pp	9%	14,915	67
-1.5 pp	13%	6,487	29.1
-2.0 pp	**18%**	**3,567**	**16**
-2.5 pp	22%	2,230	10
-3.0 pp	27%	1,511	6.8

Design check by simulation

2,000 runs resampling real Apr–Aug 2018 orders

5.4%

A/A false-positive rate (target 5%)

78.8%

power with a −2 pp effect injected (target 80%)

Both checks land close to target, so the test is sized correctly and the z-test behaves as expected on this data.

RECOMMENDATIONS

What to do, and what it is worth

1 **Start the delivery clock at payment approval**

For slow methods such as boleto, quote the delivery date from approval, or add the expected approval time to the promise shown at checkout.

~187 late orders
and ~97 1–2 star reviews avoided per year at 2017–18 volume

2 **Run the proactive delay-alert test**

Score each order at approval and randomize the top 30% by risk. Ship only if 1–2 star reviews fall and no guardrail worsens.

7,134 orders
about 16 weeks here; days at a large payments processor

3 **Put at-risk sellers on a fulfilment plan**

Share the seller scorecard, set late-rate targets, and review the 172 sellers monthly.

~862 late orders
and ~446 bad reviews over 20 months, protecting R$2.3M GMV

4 **Flag “delivered but not received” complaints**

Route them to proof-of-delivery checks before they become disputes, and track the rate by carrier and seller.

26.5%
of bad reviews on “on-time” orders describe a delivery problem

WHY THIS MATTERS IN CANADA

Canadians made **22.5 billion payments worth $12.2 trillion** in 2024, and **60%** bought something online in a given month; online purchases are expected to reach **$96.7 billion by 2028**. As the Real-Time Rail arrives (49% of Canadians find real-time payments appealing), merchants will mix instant and slower bank-rail payments at checkout. That is when a payment-aware delivery promise pays off. Sources: Payments Canada 2025 report [3], PDF pp. 7, 16, 18.

Sizing uses the controlled effects above and treats them as causal; they are observational, which is why recommendation 2 is a randomized test.

HOW IT WAS BUILT

Pipeline, metric layer and data checks

Raw
8 CSV tables loaded to SQLite

Staging
types cast, values standardized, reviews de-duplicated with window functions

Marts
fct_orders, dim_customers, dim_sellers; one row per order guaranteed

Metric registry
12 metrics in YAML, compiled to SQL

Outputs
analysis, model, test design, dashboard, this report

Every metric is defined once and compiled to SQL, so the dashboard, model and test design cannot disagree on what “late” or “GMV” means. All 14 data-quality checks run on every build: blocking checks passed, and 8 data issues were found and handled in staging. The full pipeline re-runs from raw files with identical results.

Data-quality checks

PASS = no issue · HANDLED = issue found and fixed in staging · a blocking failure would stop the pipeline

CHECK	LAYER	ROWS FLAGGED	OF	STATUS
orders pk unique	raw	0	99,441	PASS
payments type not defined	raw	3	103,886	HANDLED
payments zero installments	raw	2	103,886	HANDLED
reviews duplicate review id	raw	814	99,224	HANDLED
reviews multiple per order	raw	551	99,224	HANDLED
orders without items	raw	775	99,441	HANDLED
orders without payment	raw	1	99,441	HANDLED
items orphan seller	raw	0	112,650	PASS
orders orphan customer	raw	0	99,441	PASS
delivered missing timestamp	raw	8	99,441	HANDLED
delivered before purchase	raw	0	99,441	PASS
approved before purchase	raw	0	99,441	PASS
mart orders row count	mart	0	99,441	PASS
recon payment vs order value	mart	249	98,665	HANDLED

Limits

- Effects are observational, with controls; the test is the proof step.
- One Brazilian marketplace, 2017–18; the Canadian link is by analogy.
- Review themes use keyword rules checked on a small held-out sample.
- The experiment is designed, not run.

AI use

Claude helped draft SQL, pipeline code and the Portuguese keyword rules, and sped up debugging. Every figure in this report is computed by the pipeline from the source tables [1], and the keyword rules were evaluated on a hand-labelled held-out sample before use. Statistical and modelling libraries: [5]–[7].

SOURCES

Sources and references

Numbers in brackets refer to these sources throughout the report. Web sources were accessed on October 28, 2026.

DATA

[1] Olist. *Brazilian E-Commerce Public Dataset by Olist*. Kaggle, 2018. License: CC BY-NC-SA 4.0.
https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce
Used for: every order, payment, item, review, customer and seller figure in this report (8 tables; 99,441 raw orders, 99,092 in the Jan 2017 – Aug 2018 analysis window).

[2] bulutttt. *olist-raw-data* (mirror of [1]). Hugging Face Datasets.
https://huggingface.co/datasets/bulutttt/olist-raw-data
Used for: downloading [1]. File sizes match the original release byte for byte and a second independent mirror. Retrieved October 28, 2026.

INDUSTRY CONTEXT

[3] Payments Canada. *Canadian Payment Methods and Trends Report 2025* (12th ed.). Ottawa: Payments Canada, 2025.
https://www.payments.ca/sites/default/files/PaymentsCanada_Canadian_Payment_Methods_and_Trends_2025_EN.pdf
Used for: 22.5 billion payments worth $12.2 trillion in 2024 (PDF p. 7); 60% of Canadians bought online in a given month and online purchases expected to reach $96.7 billion by 2028 (p. 16); 49% find real-time payments appealing, Real-Time Rail forthcoming (p. 18).

[4] Payments Canada. “Canada reaches $12.2 trillion in payment transactions in 2024, with credit cards accounting for 1 in 3 transactions.” Research summary, 2025.
https://www.payments.ca/insights/research/canada-reaches-122-trillion-payment-transactions-2024-credit-cards-accounting-1-3
Used for: cross-checking the headline figures taken from [3].

METHODS AND SOFTWARE

[5] Seabold, S., & Perktold, J. “statsmodels: Econometric and statistical modeling with Python.” *Proceedings of the 9th Python in Science Conference*, 2010.
https://www.statsmodels.org
Used for: OLS with fixed effects and HC1 robust standard errors, power analysis, two-proportion z-tests.

[6] Pedregosa, F., et al. “Scikit-learn: Machine Learning in Python.” *Journal of Machine Learning Research*, 12, 2825–2830, 2011.
https://scikit-learn.org
Used for: the gradient-boosted late-delivery risk model and AUC.

[7] McKinney, W. “Data Structures for Statistical Computing in Python.” *Proceedings of the 9th Python in Science Conference*, 2010 (pandas).
https://pandas.pydata.org
Used for: data loading, joins outside SQL, and table outputs.